

Set Operations in SQL

Q. Discuss the SQL Set Operations. Explain UNION, UNION ALL, INTERSECT, and EXCEPT with examples, and list the strict rules required to use them.

📌 Definition to Remember

SQL Set Operations allow you to combine the result sets of two or more independent **SELECT** queries into a single output. They are based on mathematical set theory and include **UNION**, **UNION ALL**, **INTERSECT**, and **EXCEPT/MINUS**. All require **Union Compatibility** between the queries.

1. Rules for Set Operations (Union Compatibility)

Both **SELECT** queries must satisfy:

1. **Same Number of Columns:** Both queries must return the same number of columns.
2. **Compatible Data Types:** Corresponding columns must have compatible data types in the same order.

2. Types of Set Operations

1. UNION

- Combines results of two queries and **automatically removes duplicate rows**.

```
SELECT City FROM Customers
UNION
SELECT City FROM Suppliers;
```

Result: All unique cities from both tables.

2. UNION ALL

- Combines results but **retains all duplicates**. Significantly **faster** than **UNION** (no sorting/deduplication step).

```
SELECT City FROM Customers
UNION ALL
SELECT City FROM Suppliers;
```

Result: All city entries, including repeated cities.

3. INTERSECT

- Returns only rows **common to both** queries.

```
SELECT City FROM Customers
INTERSECT
SELECT City FROM Suppliers;
```

Result: Only cities with both a Customer AND a Supplier.

4. EXCEPT / MINUS

- Returns rows from the **first query that are NOT in the second**.
- Oracle uses **MINUS**; SQL Server/PostgreSQL use **EXCEPT**.

```
SELECT City FROM Customers
EXCEPT
SELECT City FROM Suppliers;
```

Result: Cities with Customers but NO Suppliers.

3. Quick Comparison

Operation	Duplicates	Returns
UNION	Removed	All from A + All from B
UNION ALL	Kept	All from A + All from B (with duplicates)
INTERSECT	Removed	Common rows only
EXCEPT/MINUS	Removed	Rows in A not in B

★ Must-Write Points (for 10 marks)

1. Set operations combine results of two SELECT queries into a single result set.
2. **Union Compatibility** required: same number of columns AND compatible data types.
3. **UNION** : combines all rows, **removes duplicates**.
4. **UNION ALL** : combines all rows, **keeps duplicates** (faster than UNION).
5. **INTERSECT** : returns only **common rows** present in both queries.
6. **EXCEPT** (MINUS in Oracle): returns rows in first query **not found** in the second.
7. UNION ALL is preferred for performance when duplicates are acceptable.

⚡ Quick Recall

Set Ops → Union Compatibility → UNION (no duplicates) → UNION ALL (keeps duplicates, faster) → INTERSECT (common only) → EXCEPT/MINUS (A not in B)